

Multiple Choice

1. **Answer:** A

Options: A) 33.9; B) 29.8; C) 27.5; D) 31.2; E) 24.3

Note: Correct option: A - 33.9

2. **Answer:** A

Options: A) 0.4 henry; B) 0.3 henry; C) 0.2 henry; D) 0.6 henry; E) 0.7 henry

Note: Correct option: A - 0.4 henry

3. **Answer:** B

Options: A) It is increased by two times.; B) It remains the same.; C) It becomes zero.; D) It is reduced to half.; E) It is increased by eight times.

Note: Correct option: B - It remains the same.

4. **Answer:** E

Options: A) $(f / 3)$; B) $(f / 2)$; C) $(5 f / 3)$; D) $(6 f / 5)$; E) $(3 f / 4)$

Note: Correct option: E - $(3 f / 4)$

5. **Answer:** A

Options: A) because Mars' axis is less tilted than the Earth's; B) because Mars has a smaller size than the Earth; C) because Mars has a slower rotation speed than the Earth; D) because Mars has more carbon dioxide in its atmosphere than the Earth; E) because Mars has a shorter year than the Earth

Note: Correct option: A - because Mars' axis is less tilted than the Earth's

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 0.5 cal/K/deg. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 0.245 N. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key